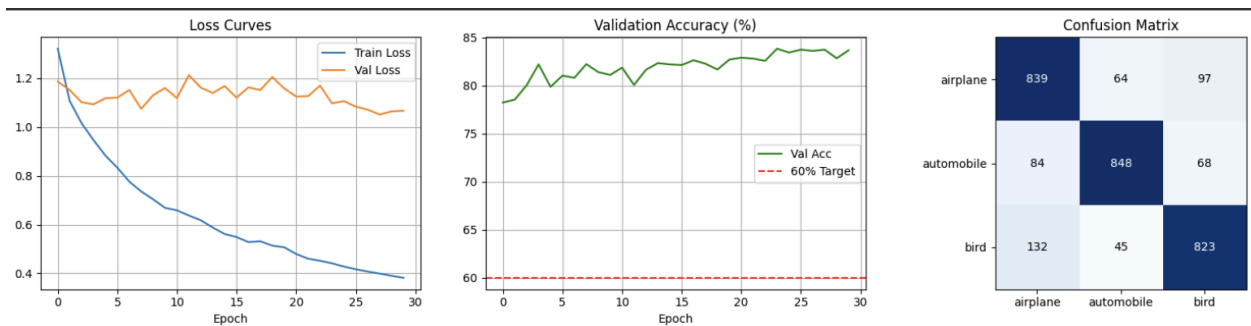


Model Performance Report

The model implemented is a 3 layer Multi-Layer Perceptron neural network. It implements 3-class filtering into Airplane, Automobile and Bird classes from the remaining 7. After flattening, formatting and normalization, the images are sent to the forward processing network (3 layer MLP). Finally using backpropagation, the losses are calculated and weights are altered for the following epoch, so that the model learns from the previous iteration.

Class	Precision	Recall	F1-Score
airplane	0.7953	0.8390	0.8165
automobile	0.8861	0.8480	0.8666
bird	0.8330	0.8230	0.8280
Overall Accuracy: 83.67%			

The neural network achieved an overall test accuracy of 83.67%. Automobile class received the highest F1-Score, most probably because feature extraction against a single RGB-value road is simpler during classification. Airplane and bird classes however exhibited high class confusion values. This may be seen as both classes display similar background textures (of the sky) and each object could have been confused with the other.



The optimisation criteria tested was SGD with momentum. Loss curves showed gradual descent, validating the optimisation criteria. Training loss decreased from 1.32 to 0.38 over 30 epochs, while validation loss dropped to 1.05. Finally accuracy stabilised at 83.67%, showing that the model achieved above average (60% baseline) metrics in accuracy.